CO1

Assessment - 2

Concept Mapping.

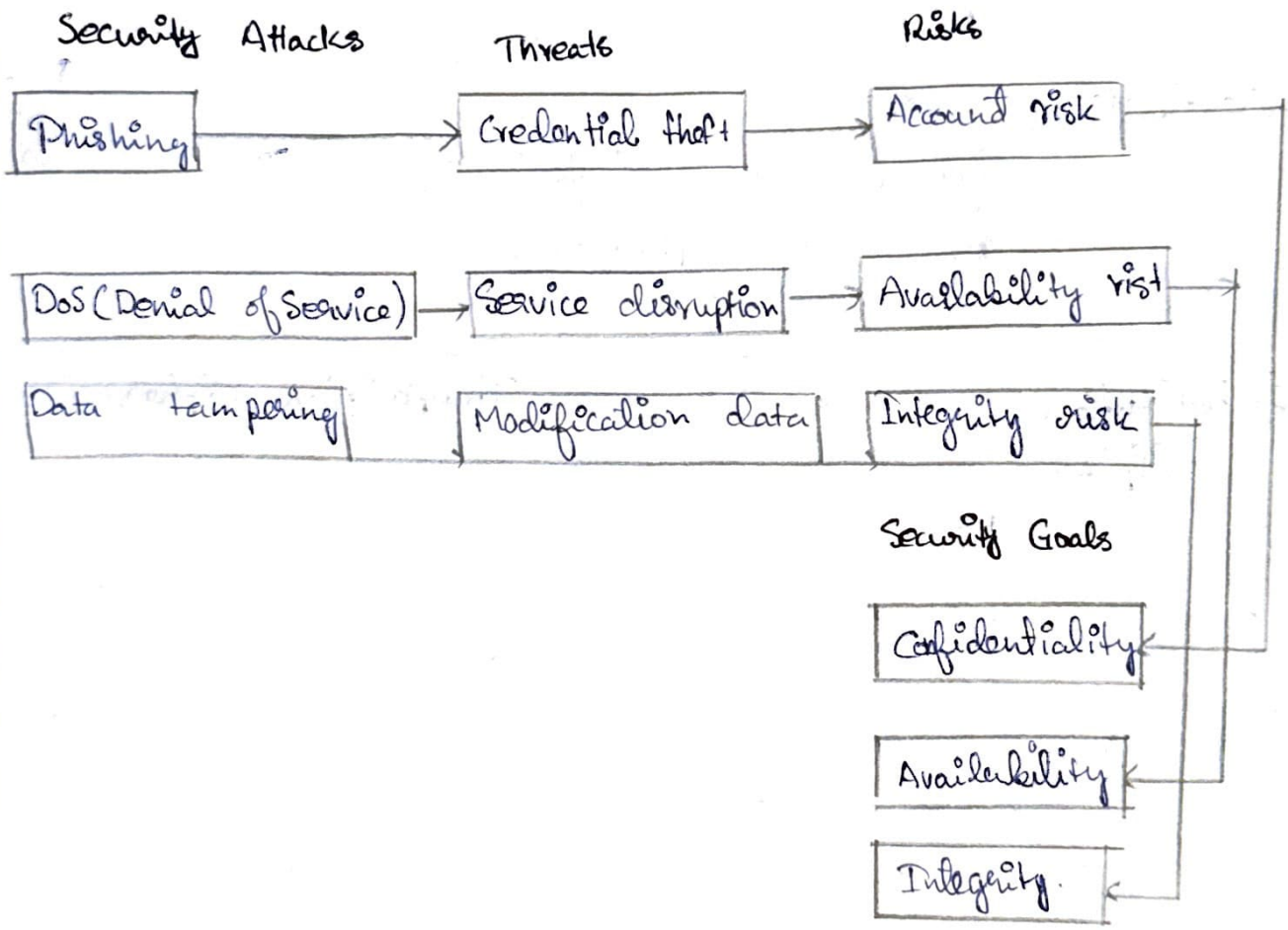
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CSA 5111

Cryptography and Network Security

1)



2)

Cipher	Key	Complexity	Strength Security
Caesar	Shift value	Low	Weak
Monoalphabetic	Substitution table	Medium	moderate
Vigenère	Keyword (multiple shifts)	Higher	Strong
Hill	Key Matrix	Higher	Strong

Conclusion:
Vigenere and Hill ciphers are stronger than Caesar & Monoalphabetic.

3)

Substitution Cipher	Transposition Cipher
-> Changes the Symbol	-> Changes the position of letter
-> Example: Caesar	-> Example: Rail fence
-> Affects the data Content	-> Keep the symbols same
-> Security level: Moderate to high	-> Security level: generally low than Substitution

Conclusion:
Both are classical encryption that transform Plaintext but Substitution changes symbols while transposition changes the position.

4)

Data Hiding

-> Hides information inside another file (eg image)

↓

Watermarking

-> A special type of data hiding used for copyrights protection.

↓

Survivability.

-> The hidden data should remain intact. (e.g. after compression)

↓

Real - world Application

-> Hidden message in an image

-> Copyright watermark

-> Robust watermark surviving compression